ab initio / classical MD

VASP, ABINIT, QBox, LAMMPS

VaspParser.py
AbinitParser.py
QBoxParser.py
LAMMPSParser2umd.py -f -i -s

repair_umd.py -f

check_umd.py -f

.check.

QUALITY OF THE RUN

INPUT/OUTPUT

.umd.

umd2out.py -f -r -s -i

.xyz.

POSCAR

LAMMPS

DeepMD

new structures

build_umd.py -f -i -s -a
--supercell --shuffle --insert
--scale --replace --remove

gofr_umd.py -f -d -s -i -c -a

.gofr.

INTERATOMIC BONDING

analyze_gofr.py -f -c -a -v

.maximum bonds.

.coordinations.

COORDINATION NUMBERS

bonding_umd.py -f -c -l -s -i

.bonding.

speciation_and_angles.py
-f -b -c -a -r -m -s -t -l

.step.

.popul.

CHEMICAL SPECIATION

.stat.

.angles.

ANGLES

.bondslives.

BOND LIFETIMES

lifetimes.py

LIFETIMES OF SPECIES

liquid_and_vapor_speciation.py
stat-concentrate.py

LIQUID / VAPOR PARTITIONING

umd_workflow.py -g : the whole panel

STRUCTURAL PROPERTIES

viscosity_umd.py
-f -i -n -c -v

.stress.scf.

.visc.

VISCOSITY

msd_umd.py -f -i -s -m -a
-n -b -e

.msd.

.diffusion.

ATOMIC DIFFUSION
(and tensor)

analyze_msd.py

msd_clusters_umd.py -f -p

.msd.

DIFFUSION OF CHEMICAL SPECIES

analyze_msd_clusters.py

umd_workflow.py -t : the whole panel

TRANSPORT PROPERTIES

averages.py -f -p -a

fullthermo.

THERMO-DYNAMICS
(P, T, energy, heat capacity)

vibrations_umd.py
-f -a -t -w

.vels.scf.

.vibr.

VIBRATIONAL SPECTRUM

vdos_thermo.py -f -t

.vdos.thermo.

.2pt.

THERMO-DYNAMICS
(entropy, free energy: harmonic, 2PT, quantum corrections)

.umd.

runs at several λ

thermo_integration_umd.py
-f -t -k -v

thermo_integration.

FREE ENERGY, ENTROPY

umd_workflow.py -d : the whole panel

THERMODYNAMIC PROPERTIES